

AJAY KANNAN

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EDUCATION

Master of Science in Computer Science, Arizona State University, Tempe

Jan 2021 – May 2023

Bachelor of Science in Computer Science, Anna University, Chennai

Aug 2014 – May 2019

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, CUDA C++, C#, JavaScript, SQL, PostgreSQL, Node.js, Bash

ML/DL Frameworks: PyTorch, TensorFlow, JAX, Huggingface, Keras, NumPy, Scikit-learn, Pandas

LLM & Agentic AI: LangChain, LlamaIndex, CrewAI, LangSmith, AWS Bedrock Agents, AWS Strands Agents, RAG Pipelines, Prompt Engineering

Vector Databases: Pinecone, Weaviate, Qdrant

Cloud & AI Services: AWS (Bedrock, SageMaker, Lambda, S3, EC2), GCP (Vertex AI, BigQuery, Dataflow), Azure, NVIDIA CUDA

Backend Development: FastAPI, Spring Boot, Node.js, Express.js, REST APIs, GraphQL, Microservices, Kafka

Databases: PostgreSQL, MongoDB, MySQL, Cassandra, DynamoDB, NoSQL

DevOps & Infrastructure: Docker, Kubernetes, Git, CI/CD, Terraform basics

Others: Predictive Modeling, ML & DL Algorithms, Big Data, Decision Analytics, Exploratory Data Analysis, Computer Vision

PROFESSIONAL EXPERIENCE

ML Systems / Backend Infrastructure Engineer | Pacific Gas & Electric, Santa Clara, CA
Present

Jul 2023 –

- Designed and deployed a production data platform (Electric Productivity Tracker) to process large-scale utility datasets, improving data processing throughput by 20% and reducing system errors by 15% through backend automation, monitoring, and reliability engineering across distributed systems
- Engineered scalable, fault-tolerant data pipelines integrating heterogeneous sources (GIS, PostgreSQL, internal services), ensuring consistent data flow and low-latency access while maintaining high system availability and processing continuous data streams
- Built automated backend validation and testing infrastructure using Python services and REST interfaces, reducing deployment verification time by 50% and improving system reliability through comprehensive unit tests, integration tests, and end-to-end validation suites
- Implemented end-to-end data engineering workflows (ETL, validation, transformation) across distributed systems, improving data integrity and reducing manual operational overhead through automated pipeline orchestration and quality checks
- Applied ML/Generative AI techniques in production systems to improve anomaly detection reliability, reducing false positives by 30% through model-assisted signal validation, statistical inference, and system-level monitoring with continuous feedback loops
- Developed observability and monitoring systems (health checks, dashboards, alerting) across servers and databases, reducing issue detection and resolution time by 50% through proactive alerting, comprehensive metrics collection, and centralized logging infrastructure
- Collaborated with cross-functional teams (engineering, operations, GIS specialists, product managers), translating technical requirements into algorithm specifications and creating comprehensive documentation including architecture diagrams, API specifications, and operational runbooks

Data Scientist | Sigtuple Inc., Bangalore, India

Jul 2021 – Dec 2021

- Applied advanced image processing techniques to biological data, utilizing Siamese Networks, GANs, and Neural Style Transfer for medical image analytics. Implemented sophisticated algorithms for automated histology image analysis using PyTorch
- Utilized alpha blending techniques to recreate histology slide images, accurately representing biological particles and membranes through innovative image processing and registration methods
- Achieved 94% accuracy enhancement in image blending efficiency through innovative generative architectures, demonstrating expertise in applying deep learning to medical imaging challenges and biological data analysis
- Built production-quality software in Python following object-oriented design principles with comprehensive testing frameworks, collaborating with domain experts to validate algorithm outputs and ensure clinical relevance

Software Engineering Intern | Microsoft, Hyderabad, India

Summer 2017 & 2018

- Identified key sensor data patterns and trends to significantly enhance crop yields through data processing and visualization using Python, ML, REST API, and PyTorch as part of the FarmBeats agricultural intelligence platform
- Incorporated a GIS-based heat-map generation system for sensor values in FarmBeats, designing algorithms for real-time

- spatial visualization of multi-sensor data (soil moisture, temperature, humidity) across agricultural fields
- Led the Shopping on Cortana project, integrating Cortana voice assistant with a shopping platform to improve user experience through natural language interaction and personalized recommendations
 - Handled backend development with Cassandra DB and C# as part of the Foundry Team in MS-IDC, developing RESTful APIs for distributed data storage and high-availability service integration

PROJECTS AND RESEARCH

Few-Shot Learning for TCR-Epitope Prediction | Arizona State University Aug 2022 – May 2023

- Developed contrastive loss for Few Shot Learning in TCR-Epitope affinity prediction using PyTorch, improving accuracy by 30% over baseline methods through novel architectural improvements and loss function design
- Implemented a Siamese network to gauge the efficiency of different K-values in few-shot learning, conducting comprehensive experiments and statistical analysis to validate model performance across varying data distributions
- Presented research findings as a poster at the MORE symposium at ASU, communicating complex technical concepts to diverse audiences and demonstrating ability to contribute to scientific literature

Gesture Recognition Systems | Solarillion Foundation Aug 2015 – Dec 2018

- Designed and developed both static and dynamic American Sign Language Gesture Recognition Systems, achieving 94–97% accuracy using accelerometer sensors, Arduino, and machine learning algorithms (Extra Trees, Random Forest, Ridge Classifier)
- Implemented advanced distance metric techniques to enhance recognition performance, designing custom feature extraction methods for both single-frame static gestures and temporal sequence dynamic gestures
- Built complete software pipeline integrating sensor data acquisition, feature processing, model training, and real-time inference with object-oriented Python architecture
- Led IoT research group, managing R&D projects and mentoring junior researchers as teaching assistant, fostering cross-disciplinary innovation and scientific methodology
- Published research in IEEE Global Internet of Things Summit 2017 and ArXiv, demonstrating ability to conduct rigorous research and communicate findings to scientific community

Chatbot with LM Studio

- Engineered a voice-enabled on-device LLM assistant using LM Studio and DeepSeek, building real-time inference pipelines for low-latency conversational AI without cloud reliance
- Designed API orchestration architecture handling speech-to-text, LLM inference, and text-to-speech components with fault-tolerant error handling
- Implemented privacy-preserving conversational AI by running models locally, eliminating data transmission to external servers

Interactive Banking Chatbot

- Engineered a production-style LLM-powered banking chatbot using GPT models, LangChain, and LlamaIndex for intelligent financial query handling
- Designed backend orchestration for context management, prompt workflows, and intent handling to deliver accurate, context-aware responses across multi-turn conversations
- Implemented Retrieval-Augmented Generation (RAG) pipelines with vector databases (Pinecone) to ground LLM responses in verified banking documentation
- Built scalable API integrations and optimized NLP pipelines, improving response relevance and system reliability while delivering seamless user experience

Twitter Data Analysis

- Engineered real-time ML retrieval systems using ANN and SVR, achieving 98% accuracy in context-aware data retrieval
- Integrated streaming data pipelines with Kafka and backend services for low-latency, production-ready deployment handling high-throughput data
- Built web application interface with real-time data streaming for improved accessibility and interactive visualization of analysis results

PUBLICATIONS & ACHIEVEMENTS

- [Few Shot Learning for TCR-Epitope Binding Affinity Prediction](#), MORE Symposium, ASU, 2023
- [A. Kannan et al., "Low-cost static gesture recognition system using MEMS accelerometers," IEEE Global IoT Summit, 2017](#)
- [G. K. Gudur et al., "A Generic Multi-modal Dynamic Gesture Recognition System using Machine Learning," ArXiv abs/1809.05839, 2018](#)